

Linear Algebra

(1)

Rubric

MID-II

Q1

(a) vector equation of plane is

$$\vec{x} = x_0 + t_1 \vec{v}_1 + t_2 \vec{v}_2$$

$$\text{here } x_0 = (3, 0, -1, 2), \vec{v}_1 = (2, -4, 1, 5)$$

$$\text{and } \vec{v}_2 = (-8, 6, 0, 7)$$

$$\text{let } \vec{x} = (x_1, x_2, x_3, x_4)$$

So vector equation of plane in $\mathbb{R}^4$ is

$$\vec{x} = x_0 + t_1 \vec{v}_1 + t_2 \vec{v}_2$$

$$\Rightarrow \{(x_1, x_2, x_3, x_4) = (3, 0, -1, 2) + t_1(2, -4, 1, 5) + t_2(-8, 6, 0, 7)\}$$

which yields the parametric equations

$$x_1 = 3 + 2t_1 - 8t_2$$

$$x_2 = -4t_1 + 6t_2$$

$$x_3 = -1 + t_1$$

$$\text{and } x_4 = 2 + 5t_1 + 7t_2$$

(b) The planes $2x - y - z = 5$ and $-4x + 2y + 2z = 12$

are parallel since their normals

$(2, -1, -1)$ and $(-4, 2, 2)$ are parallel vectors.

To find the distance D between the planes, we can select an arbitrary point in one of the planes.

By setting $y = z = 0$ in the equation $-4x + 2y + 2z = 12$, we obtain a point $P_0(-3, 0, 0)$ in the plane.

(2)

So the distance between P_0 and the plane $2x - y - z = 5$ is

$$D = \frac{|2(-3) - (0) - (0) - 5|}{\sqrt{2^2 + (-1)^2 + (-1)^2}}$$

$$D = \frac{|-6 - 5|}{\sqrt{4 + 1 + 1}} = \frac{11}{\sqrt{6}}$$

$$\Rightarrow \boxed{D = \frac{11}{\sqrt{6}}}$$

Q2

$$x_1 + 3x_2 - 2x_3 + 2x_5 = 0$$

$$2x_1 + 6x_2 - 5x_3 - 2x_4 + 4x_5 - 3x_6 = 0$$

$$5x_3 + 10x_4 + 15x_6 = 0$$

$$2x_1 + 6x_2 + 8x_4 + 4x_5 + 18x_6 = 0$$

$$\begin{bmatrix} 1 & 3 & -2 & 0 & 2 & 0 \\ 2 & 6 & -5 & -2 & 4 & -3 \\ 0 & 0 & 5 & 10 & 0 & 15 \\ 2 & 6 & 0 & 8 & 4 & 18 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

The augmented matrix is

$$\left[\begin{array}{cccccc|c} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 2 & 6 & -5 & -2 & 4 & -3 & 0 \\ 0 & 0 & 5 & 10 & 0 & 15 & 0 \\ 2 & 6 & 0 & 8 & 4 & 18 & 0 \end{array} \right]$$

(3)

$$R_2 - 2R_1 \rightarrow R_2, \quad R_4 - 2R_1 \rightarrow R_4$$

$$\left[\begin{array}{cccccc|c} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 0 & 0 & -1 & -2 & 0 & -3 & 0 \\ 0 & 0 & 5 & 10 & 0 & 15 & 0 \\ 0 & 0 & 4 & 8 & 0 & 18 & 0 \end{array} \right]$$

$$(-1)R_2 \rightarrow R_2$$

$$\left[\begin{array}{cccccc|c} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 3 & 0 \\ 0 & 0 & 5 & 10 & 0 & 15 & 0 \\ 0 & 0 & 4 & 8 & 0 & 18 & 0 \end{array} \right]$$

$$R_3 - 5R_2 \rightarrow R_3, \quad R_4 - 4R_2 \rightarrow R_4$$

$$\left[\begin{array}{cccccc|c} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 6 & 0 \end{array} \right]$$

$$R_3 \leftrightarrow R_4$$

$$\left[\begin{array}{cccccc|c} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 & 6 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

$$(\frac{1}{6})R_3 \rightarrow R_3$$

$$\left[\begin{array}{cccccc|c} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

$$R_2 - 3R_3 \rightarrow R_2$$

$$\begin{bmatrix} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$R_1 + 2R_2 \rightarrow R_1$$

$$\begin{bmatrix} 1 & 3 & 0 & 4 & 2 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

which shows x_2, x_4 and x_5 are free variables.

So let $x_2 = r$, $x_4 = s$ and $x_5 = t$

Finally the general solution vector is

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix} = \begin{bmatrix} -3r-4s-2t \\ r \\ -2s \\ s \\ t \\ 0 \end{bmatrix} \quad (or) \quad (x_1, x_2, x_3, x_4, x_5, x_6) = (-3r-4s-2t, r, -2s, s, t, 0)$$

Now we have four row vectors

$$\vec{r}_1 = (1, 3, -2, 0, 2, 0)$$

$$\vec{r}_2 = (2, 6, -5, -2, 4, -3)$$

$$\vec{r}_3 = (0, 0, 5, 10, 0, 15)$$

$$\vec{r}_4 = (2, 6, 0, 8, 4, 18)$$

$$\text{Now } \vec{r}_1 \cdot \vec{x} = 1(-3r-4s-2t) + 3(r) - 2(-2s) + 0(s) + 2(t) + 0(0) = 0$$

$$\vec{r}_2 \cdot \vec{x} = 2(-3r-4s-2t) + 6(r) - 5(-2s) - 2(s) + 4(t) - 3(0) = 0$$

$$\vec{r}_3 \cdot \vec{x} = 0(-3r-4s-2t) + 0(r) + 5(-2s) + 10(s) + 0(t) + 15(0) = 0$$

$$\text{and } \vec{r}_4 \cdot \vec{x} = 2(-3r-4s-2t) + 6(r) + 0(-2s) + 8(s) + 4(t) + 18(0) = 0$$

which establishes the orthogonality.

(5)

Q3

$$(i) \quad \vec{u} + \vec{v} = (-1+3+1, 2+4+1) \\ = (3, 7)$$

$$k\vec{u} = 3(-1, 2) = (0, 6)$$

$$(ii) \quad \text{let } \vec{o} = (x, y)$$

$$\text{and } \vec{u} + \vec{o} = \vec{u}$$

$$(-1, 2) + (x, y) = (-1, 2)$$

$$(-1+x+1, 2+y+1) = (-1, 2)$$

$$(x, y+3) = (-1, 2)$$

$$\Rightarrow \begin{cases} x = -1 \\ y+3 = 2 \\ \Rightarrow y = -1 \end{cases}$$

so

$$\Rightarrow \vec{o} = (-1, -1)$$

$$\text{let } -\vec{u} = (u_1^*, u_2^*)$$

$$\text{and } \vec{u} + (-\vec{u}) = \vec{o}$$

$$(-1, 2) + (u_1^*, u_2^*) = (-1, -1)$$

$$(-1+u_1^*+1, 2+u_2^*+1) = (-1, -1)$$

$$(u_1^*, u_2^*+3) = (-1, -1)$$

$$\Rightarrow \begin{cases} u_1^* = -1 \\ u_2^* = -4 \end{cases}$$

$$\Rightarrow -\vec{u} = (-1, -4)$$

$$(iii) (a) \quad k(\vec{u} + \vec{v}) = k(u_1+u_2+1, u_2+v_2+1) \\ = (0, ku_2+kv_2+k)$$

$$k\vec{u} = k(u_1, u_2) = (0, ku_2) \quad , \quad k\vec{v} = k(v_1, v_2) = (0, kv_2)$$

$$k\vec{u} + k\vec{v} = (0, ku_2) + (0, kv_2) = (0+0+1, ku_2+kv_2+1) = (1, ku_2+kv_2+1)$$

(6)

$$\Rightarrow \left\{ k(\vec{u} + \vec{v}) \neq k\vec{u} + k\vec{v} \right\} \text{ (not hold)}$$

$$(b) \quad (k+m)\vec{u} = (k+m)(u_1, u_2) = (0, (k+m)u_2) \\ = (0, ku_2 + mu_2)$$

$$k\vec{u} = k(u_1, u_2) = (0, ku_2), \quad m\vec{u} = m(u_1, u_2) = (0, mu_2)$$

$$k\vec{u} + m\vec{u} = (0, ku_2) + (0, mu_2) \\ = (0+0+1, ku_2 + mu_2 + 1) \\ = (1, ku_2 + mu_2 + 1)$$

$$\Rightarrow \left\{ (k+m)\vec{u} \neq k\vec{u} + m\vec{u} \right\} \text{ (not hold)}$$

$$(c) \quad 1\vec{u} = 1(u_1, u_2) = (0, u_2) \neq (u_1, u_2) = \vec{u}$$

$$\Rightarrow \left\{ 1\vec{u} \neq \vec{u} \right\} \text{ (not hold)}$$

Q4

$$A = \begin{bmatrix} 1 & 4 & 5 & 6 & 9 \\ 3 & -2 & 1 & 4 & -1 \\ -1 & 0 & -1 & -2 & -1 \\ 2 & 3 & 5 & 7 & 8 \end{bmatrix}$$

$$R_2 - 3R_1 \rightarrow R_2, \quad R_3 + R_1 \rightarrow R_3, \quad R_4 - 2R_1 \rightarrow R_4$$

$$\therefore \begin{bmatrix} 1 & 4 & 5 & 6 & 9 \\ 0 & -14 & -14 & -14 & -28 \\ 0 & 4 & 4 & 4 & 8 \\ 0 & -5 & -5 & -5 & -10 \end{bmatrix}$$

$$\left(-\frac{1}{14}\right)R_2 \rightarrow R_2, \quad \left(\frac{1}{4}\right)R_3 \rightarrow R_3, \quad \left(-\frac{1}{5}\right)R_4 \rightarrow R_4$$

(7)

$$\begin{bmatrix} 1 & 4 & 5 & 6 & 9 \\ 0 & 1 & 1 & 1 & 2 \\ 0 & 1 & 1 & 1 & 2 \\ 0 & 1 & 1 & 1 & 2 \end{bmatrix}$$

$$R_3 - R_2 \rightarrow R_3, \quad R_4 - R_2 \rightarrow R_4$$

$$\begin{bmatrix} 1 & 4 & 5 & 6 & 9 \\ 0 & 1 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

→ row echelon form

$$R_1 - 4R_2 \rightarrow R_1$$

$$\begin{bmatrix} 1 & 0 & 1 & 2 & 1 \\ 0 & 1 & 1 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

→ reduced row echelon form

The general solution of the system is
 $(x_1, x_2, x_3, x_4, x_5) = (-r - 2s - t, -r - s - 2t, r, s, t)$

(or)

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = r \begin{bmatrix} -1 \\ -1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + s \begin{bmatrix} -2 \\ -1 \\ 0 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ -2 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

(1)

Therefore

Basis for the null space of $A = \begin{bmatrix} -1 \\ -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ -1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} -1 \\ -2 \\ 0 \\ 0 \\ 1 \end{bmatrix}$

Basis for the column space of $A = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 4 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$

Basis for the row space of $A = [1 \ 4 \ 5 \ 6 \ 9]$
 and $[0 \ 1 \ 1 \ 1 \ 2]$

⑧

(ii) $\text{rank}(A) = \text{number of leading variables}$
 $= 2$

$$\Rightarrow \boxed{\text{rank}(A) = 2}$$

$\text{Nullity}(A) = \text{number of free variables}$
 $= 3$

$$\Rightarrow \boxed{\text{nullity}(A) = 3}$$
